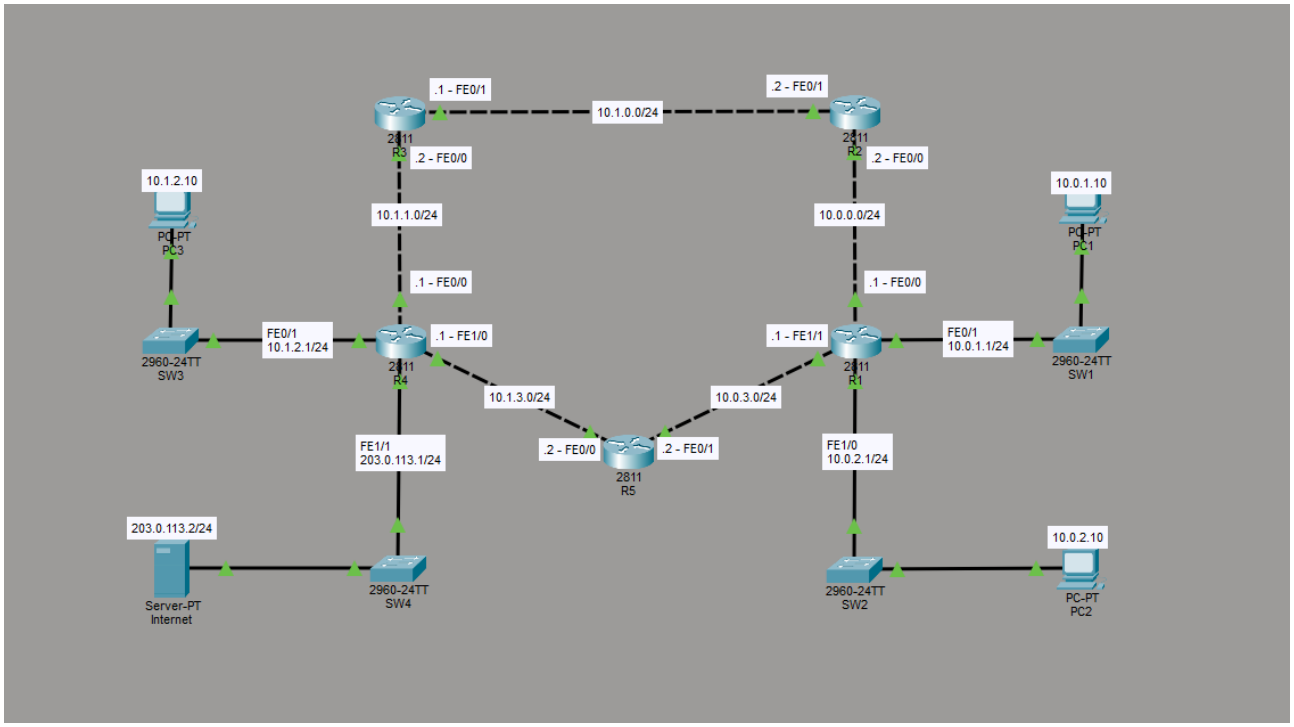
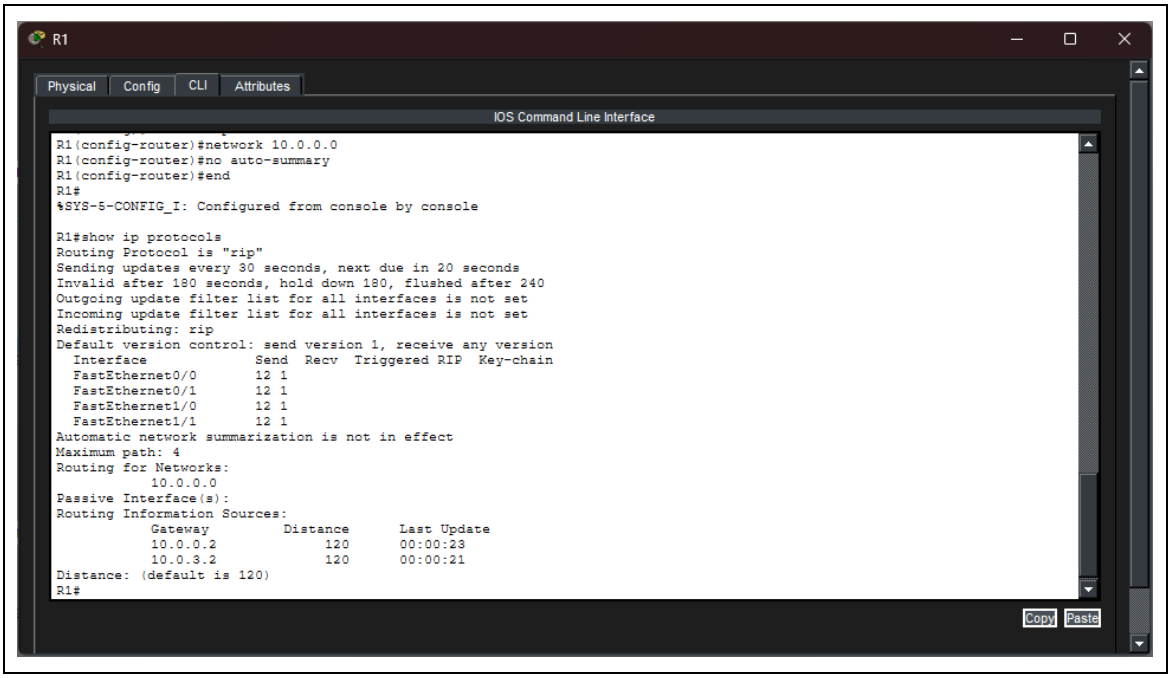
NET512 — Weekly Practical

NAME	Christiaan van Wyk	WEEK	Week 8
STUDENT NO.	20261193	DATE	25 September 2026

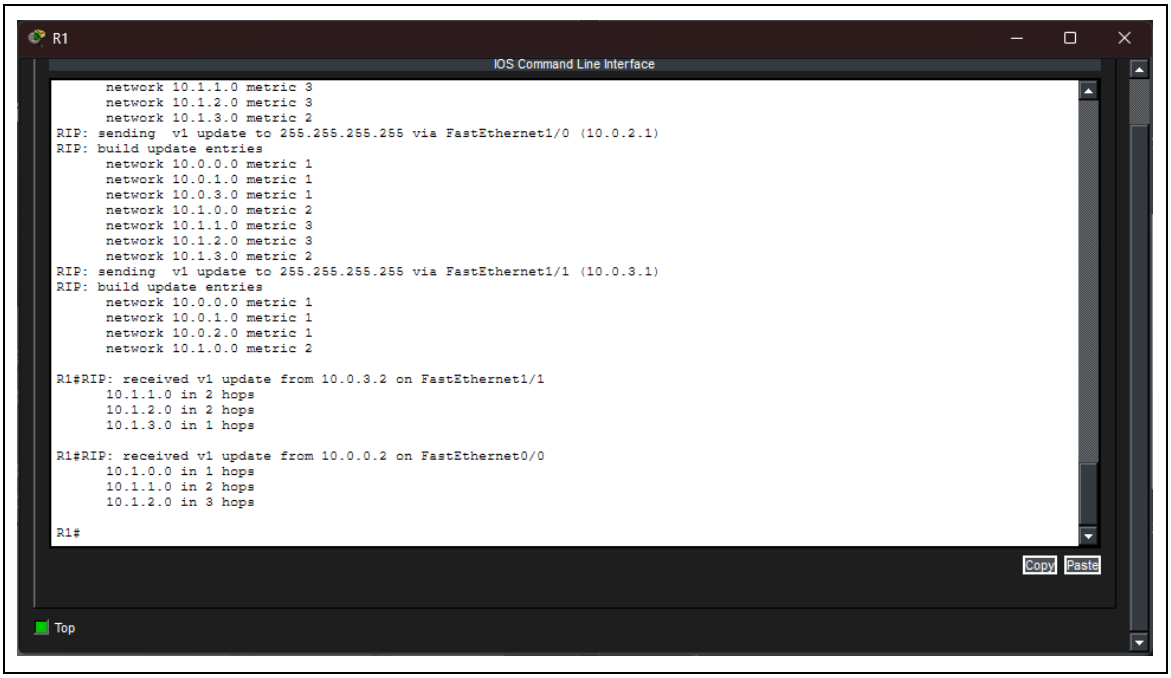
Topology



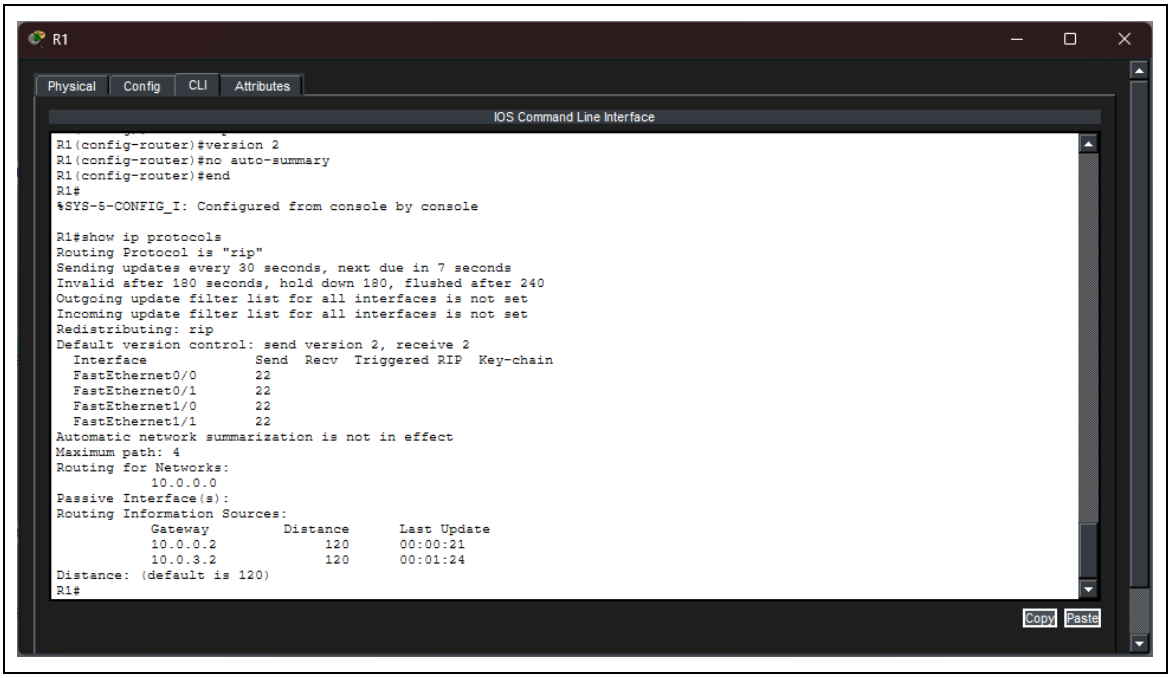
Task 74 — RIP configuration



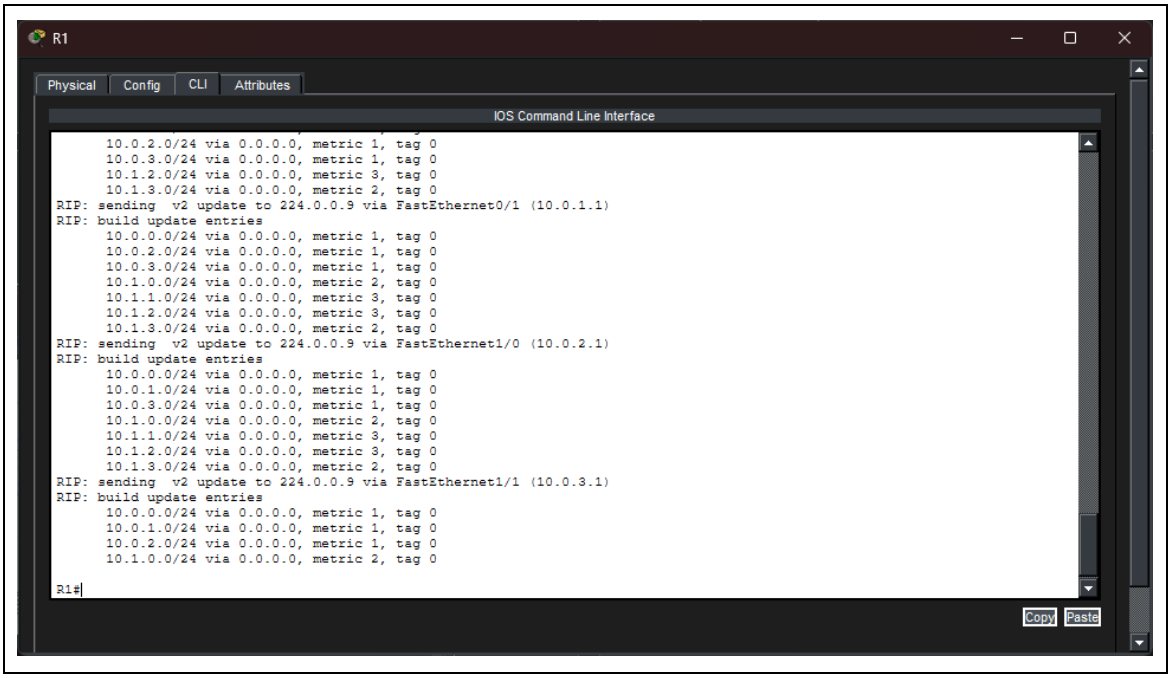
Task 75 — RIP updates & Traffic type



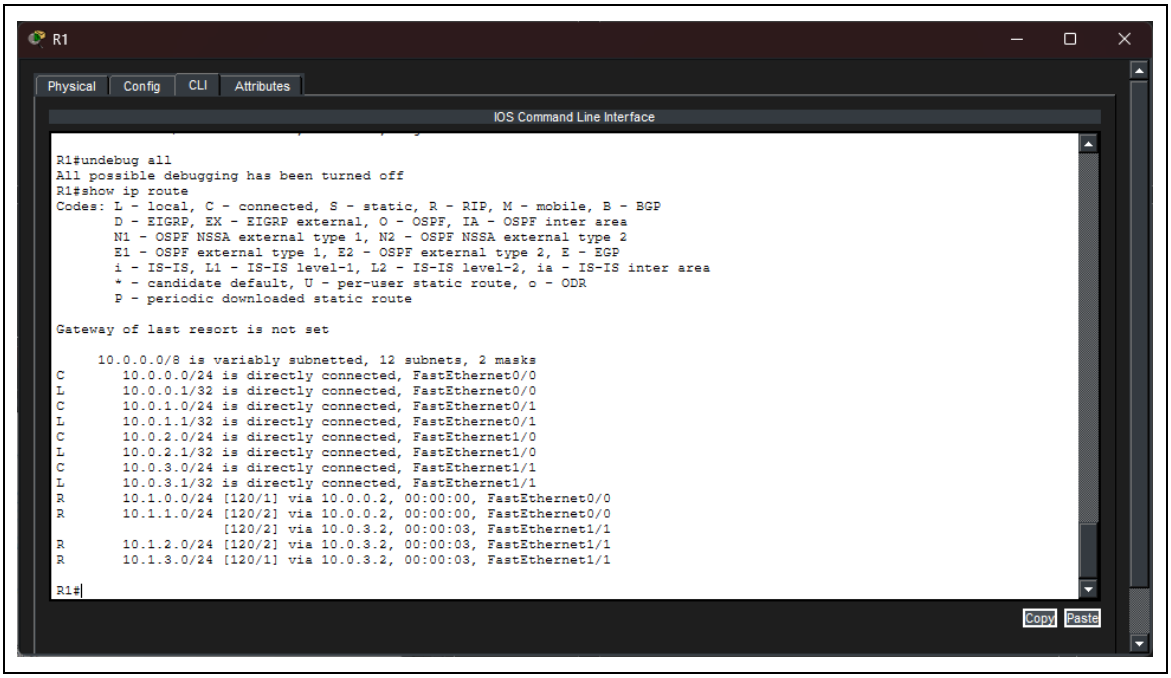
Task 76 – Upgrade to v2



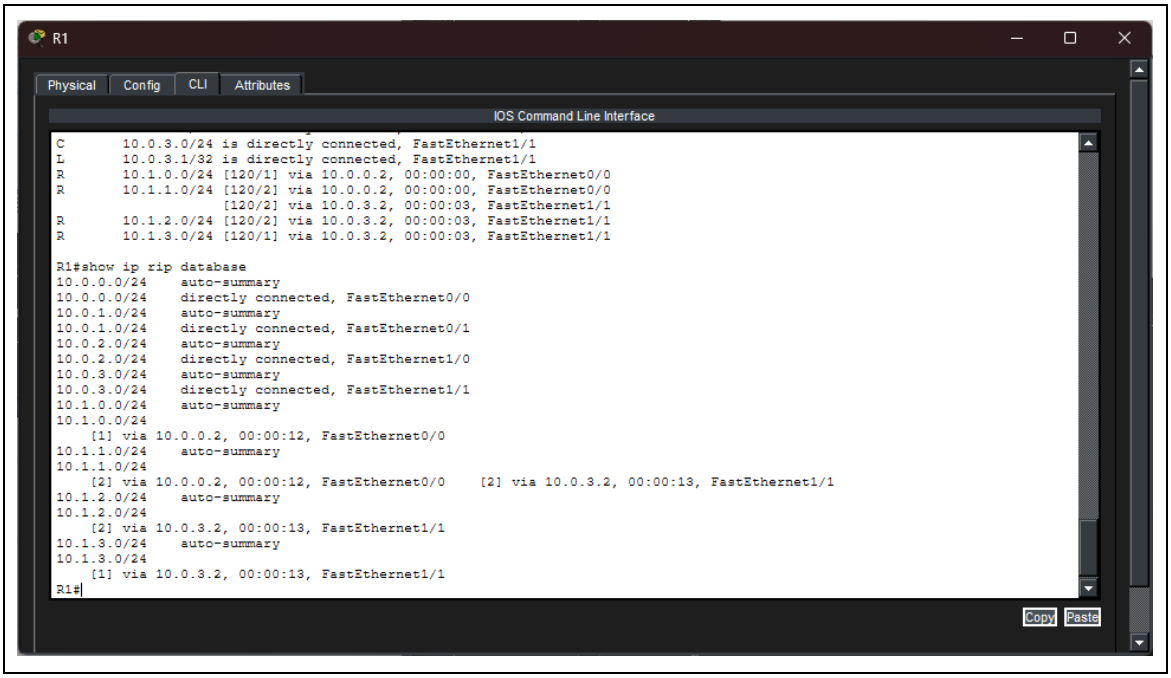
Task 77 — Traffic Type Observation



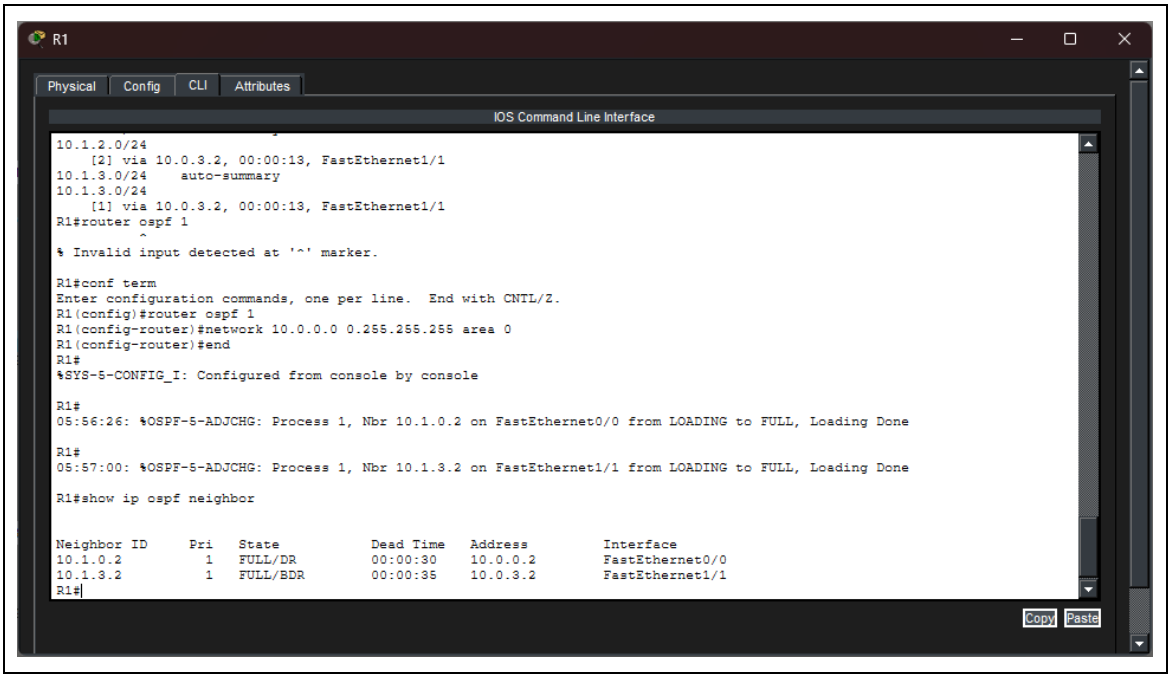
Task 78 — Learned Route



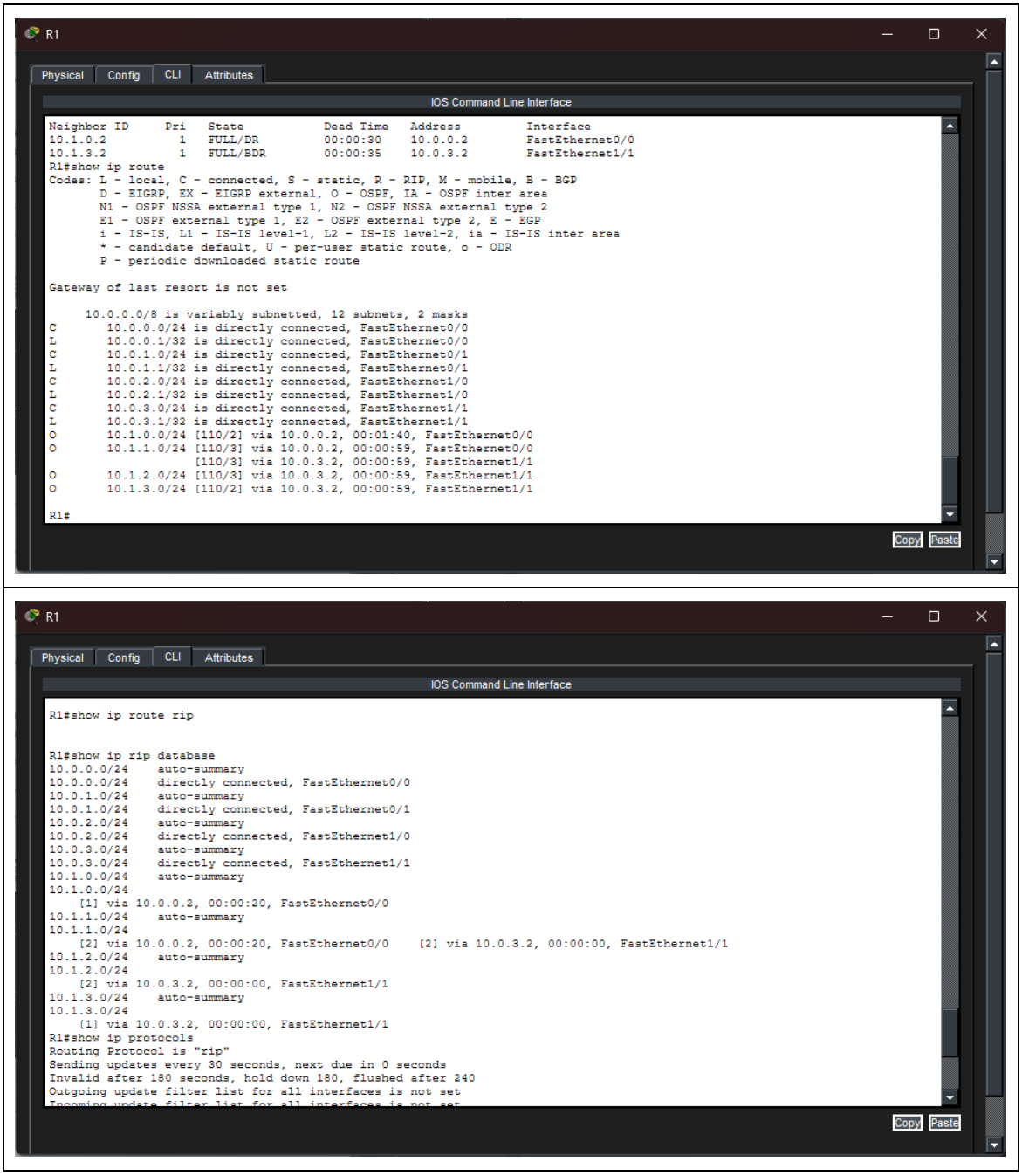
Task 79 — RIP database



Task 80 — OSPF configuration



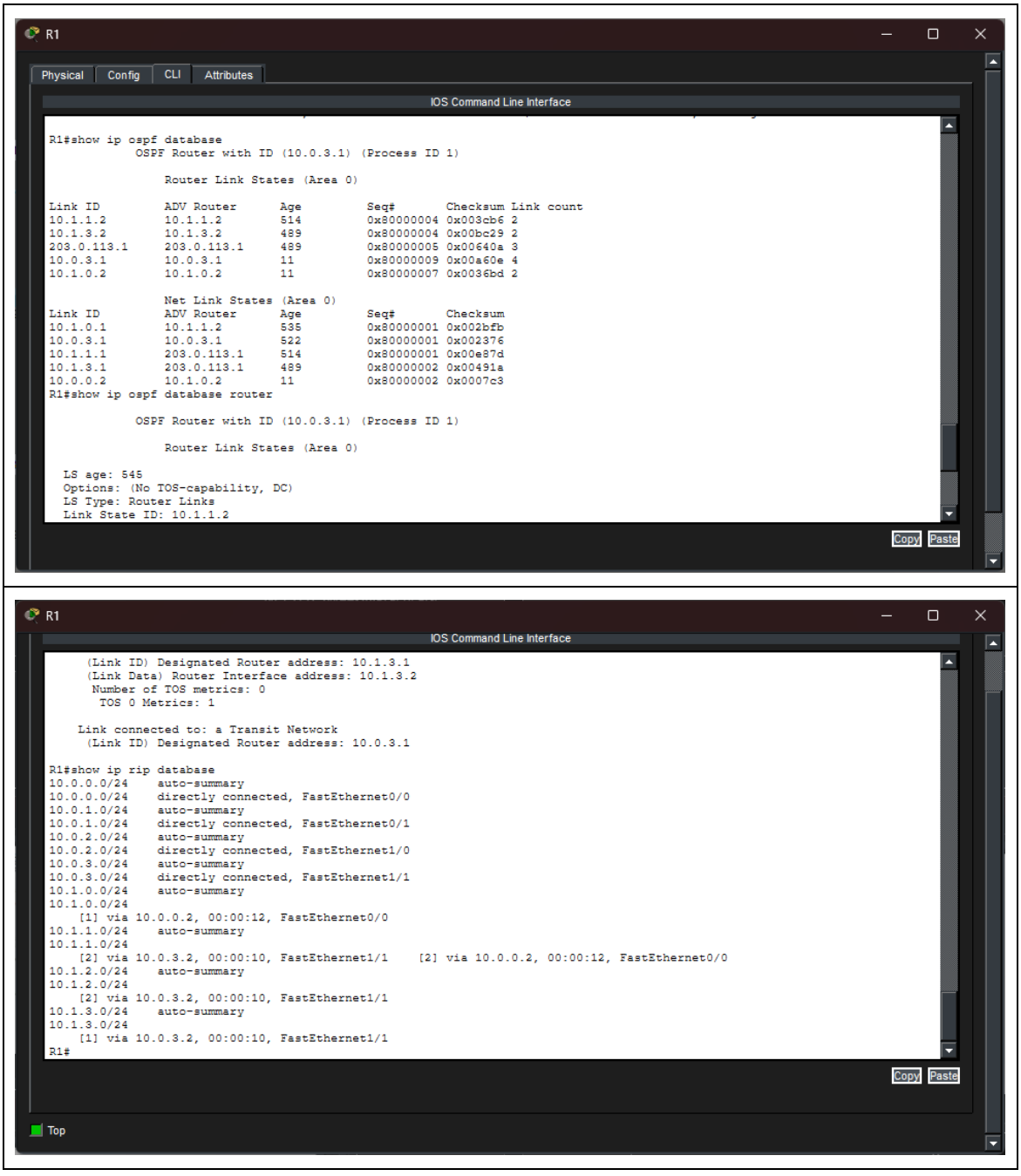
Task 81 — OSPF converging



Task 82 — Disabling interface



Task 83 — OSPF database



Reflection Questions

Task 84 – Traffic
RIPv1 makes use of broadcast traffic for updates, compared to v2 which changes this from broadcast to multicast changing it from sending to every device on the network/segment to only the reserved group address (only to joined devices).

Task 85 – OSPF configuration
There were 2 routes configured before OSPF because RIP found 2 paths that were cost equal and thus installed both in a load-sharing capacity. Both paths take 2 hops so neither is better than the other.

Task 86 – RIP route presence
Once OSPF converges RIP routes are no longer present in the routing table, they have been replaced by the OSPF routes instead. They are still there, inside the running rip database just no longer in the routing table. The reason for this is that because both RIP and OSPF are providing routes to the same destinations the IOS compares their administrative distance, and since OSPF has an AD of 110 and RIP has 120, OSPF is considered the better option and used.

Additionally, the reason for there now only being a single path to 10.1.1.0/24 is because OSPF does not count hops like RIP. Instead, it creates a sum of the cost of each outgoing interface based on bandwidth. And compared to hops this reflects the actual capacity of each link with each having a different cost through Dijkstra's algorithm. The one with the lowest cost wins.

Task 87 – Routing Table changes
After the interface was disabled multiple changes occurred. The outgoing interface of R1 switched towards the one facing R5, the OSPF cost of the R5 route changed compared to the one R2 had and lastly the OSPF neighbour table lost an adjacent neighbour and only displayed one instead of two.

Task 88 – RIP/OSPF key difference
The key difference between the two databases is that the RIP one is a list of destinations whereas the OSPF one is a map of the network. One lists destinations, their hop counts and who advertised them. While the other returns advertisements from all routers in the area, describing their own links and costs.

This difference is there because the two use different algorithm families in their functionality. RIP is distance-vector, only telling neighbours your conclusions, not your connections. OSPF is link-state, a detailed breakdown of all you direct links, raw information instead of conclusions.